CHAPTER

11 Chemical Kinetics

For NSEJS 2025

RATE OF REACTION & RATE LAW

In the thermodynamics, we have studied whether a reaction will take place or not and if it does then upto what extent (chemical equilibrium), In this chapter we will study about how fast a chemical reaction takes place and what are the different factors affecting this rate of chemical reaction. How to optimise the conditions as to maximise the output in optimum time. The last part of chapter will be dealing with the mechanism of a chemical reaction and catalysis.

(a) Rate/Velocity of Chemical Reaction

The rate of change of concentration with time of different chemical species taking part in a chemical reaction is known as **rate of reaction of that species.**

$$\text{Rate} = \frac{\Delta c}{\Delta t} = \frac{\text{mol}/\text{lit.}}{\text{sec}} = \text{mol lit}^{-1}\,\text{time}^{-1} = \text{mol dm}^{-3}\,\text{time}^{-1}$$

Rate is always defined in such a manner so that it is always a positive quantity.

(b) Types of Rates of Chemical Reaction

For a reaction $R \longrightarrow P$

$$\textbf{Average rate} = \frac{\text{Total change in concentration}}{\text{Total time taken}}$$

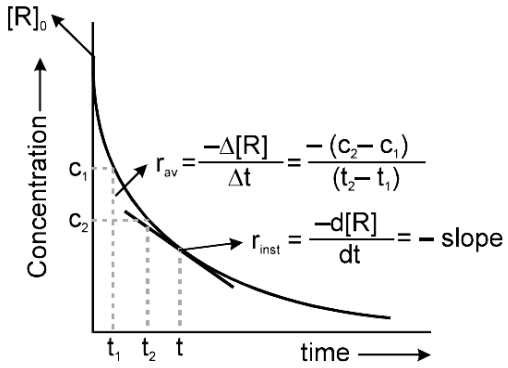
$$= \frac{\Delta c}{\Delta t} = -\frac{\Delta[R]}{\Delta t} = \frac{\Delta[P]}{\Delta t}$$

Instantaneous rate: rate of reaction at a particular instant.

$$R_{\text{instantaneous}} = \lim_{t \to 0}\left[\frac{\Delta c}{\Delta t}\right] = \frac{dc}{dt} = -\frac{d[R]}{dt} = \frac{d[P]}{dt}$$

Instantaneous rate can be determined by drawing a tangent at time t on curve drawn for concentration versus time.

Initial Rate: Instantaneous rate at 't = 0' is called initial rate [slope of tangent at t = 0].



Relation between reaction rates of different species involved in a reaction:

For the reaction: $N_2 + 3H_2 \longrightarrow 2NH_3$

Rate of reaction of $N_2 = -\frac{d[N_2]}{dt}$

Rate of reaction of $H_2 = -\frac{d[H_2]}{dt}$

Rate of reaction of $NH_3 = \frac{d[NH_3]}{dt}$

These rates are not all equal. Therefore by convention the rate of a reaction is defined as

Rate of reaction $= -\frac{d[N_2]}{dt} = -\frac{1d[H_2]}{3dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$

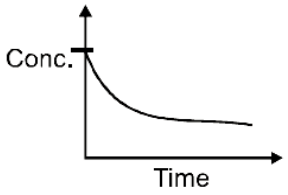
Note: Rate of reaction value is dependent on the stoichiometric coefficients used in the reaction while rate of any species will be fixed value under given conditions.

(c) Factor Affecting Rate of Chemical Reaction

(i) Concentration (ii) Nature of reactants & products

(iii) Catalyst (iv) Temperature

(i) Concentration: We known from law of mass action that Rate is proportional to concentration of reactants. "So, generally rate of reaction decreases with passage of time, since concentration of reactants decreases.



(ii) Nature of reactants & Products:

(a) Physical state of reactants:

Gaseous state > Liquid state > Solid state

Decreasing order of rate of reaction.

(b) Physical size of reactants: As we decreases the particle size rate of reaction increases since surface area increases.

(c) Chemical nature of reactants:

- If more bonds are to be broken, the rate of reaction will be slow.
- Similarly bond strength is more, rate of reaction will be slow.

(iii) Catalyst:

- Presence of positive catalyst lower down the activation energy hence increases the rate of reaction.
- presence of negative catalyst increases activation energy hence decreases the rate of reaction.

(d) RATE LAW (DEPENDENCE OF RATE ON CONCENTRATION OF REACTANTS)

The representation of rate of reaction in terms of the concentration of the reactants is called the rate law. It can only be established by experiments.

Generally rate law expressions are not simple and these may differ for the same reaction on conditions under which the reaction is being carried out.

But for large number of reactions starting with pure reactants we can obtain simple rate laws. For these reactions:

$$\textbf{Rate} \propto \textbf{(conc.)}^{\textbf{order}}$$

$$\textbf{\textit{Rate = K (conc.)}}^{\textbf{\textit{order}}} \textbf{\textit{ (differential rate equation or rate expression)}}$$

Where K = Rate constant = specific reaction rate = rate of reaction when concentration is unity unit of $K = (\text{conc})^{1-\text{ order}}\,\text{time}^{-1}$

Note: Value of K is a constant for a given reaction, depends only on temperature.

Solved Examples

Example: In a reaction, the concentration of a reactant (A) changes from 0.200 mol litre^{-1} in 10 minutes. What is the average rate of reaction during this interval?

Solution: $\Delta(A) = [A]_{final} - [A]_{initial} = [0.150 - 0.200] = -0.050$ mol litre^{-1}

$\Delta t = 10$ minutes

ORDER OF REACTIONS

Let there be a reaction

$m_1A + m_2B \longrightarrow$ products.

Now, if on the basis of experiment, we find that

$R \propto [A]^p [B]^q$ Where p may or may not be equal to m_1 & similarly q may or may not be equal to m_2.

p is order of reaction with respect to reactant A and q is order of reaction with respect to reactant B and (p + q) is **overall order of the reaction.**

Note: Order of a reaction can be 'zero' or any whole number, can be a fractional number and it can even be negative with respect to a particular reactant. But overall order is not found to be negative for any reaction till observed.

Examples showing different values of order of reactions:

Reaction	Rate law	Order
$2N_2O_5(g) \longrightarrow 4NO_2(g) + O_2(g)$	$R = K[N_2O_5]^1$	1
$5Br^-(aq)+BrO_3^-(aq)+6H^+(aq) \longrightarrow 3Br_2(\ell)+3H_2O(\ell)$	$R= K[Br^-][BrO_3^-][H^+]^2$	1 + 1 + 2 = 4
$H_2(Para) \longrightarrow H_2(ortho)$	$R = K[H_{2\,(Para)}]^{3/2}$	3/2
$NO_2(g) + CO(g) \longrightarrow NO(g) + CO_2(g)$	$R = K[NO_2]^2[CO]^o$	2 + 0 = 2
$2O_3(g) \longrightarrow 3O_2(g)$	$R = K[O_3]^2[O_2]^{-1}$	2 – 1 = 1
$H_2 + Cl_2 \xrightarrow{hv} 2HCl$	$R = K[H_2]^0[Cl_2]^0$	0 + 0 = 0

The reaction (B) does not take place in one single step. It is almost impossible for all the 12 molecules of the reactants to be in a state of encounter simultaneously. Such a reaction is called **complex reaction** and takes places in a sequence of a number of **elementary reactions**. For an elementary reaction the sum of stoichiometric coefficients = order of the reactions. But for complex reactions order is to be experimentally calculated.

Solved Examples

Example: The reaction $2NO(g) + Cl_2(g) \longrightarrow 2NOCl(g)$ is second order in NO and first order in Cl_2. In a volume of 2 dm^3, 5 mole of nitric oxide and 2 mol of Cl_2 were brought together, and the initial rate was 2.4×10^{-3} mole $dm^{-3}\,s^{-1}$. What will be the rate when half of the chlorine has reacted?

Solution: $2NO + Cl_2 \longrightarrow 2NaCl$

5 2

$$R = k\left(\frac{5}{2}\right)^2$$

$$2.4 \times 10^{-3} = k\left(\frac{15}{4}\right)$$

$$R = \frac{4 \times 2.4 \times 10^{-3}}{25}$$

$$R = \frac{4 \times 2.4 \times 10^{-3}}{25}\left[\frac{3}{2}\right]^2\left[\frac{1}{2}\right]$$

$$R = \frac{4 \times 2.4 \times 10^{-3}}{25} \times \frac{9 \times 1}{8} = 4.32 \times 10^{-4}\ M\ sec^{-1}$$

(a) Integrated Rate Law

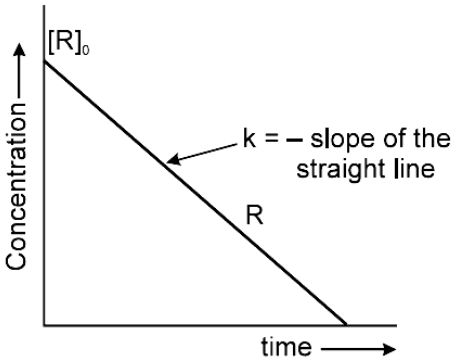
(i) Zero order reaction:

For a zero order reaction

General rate law is, Rate = k [conc.]^o = constant

If C_0 is the initial concentration of a reactant and C_t is the concentration at time 't' then

Rate = k = $\frac{C_0 - C_t}{'t'}$ or $kt = C_0 - C_t$ or $C_t = C_0 - kt$



Unit of K = same as that of Rate = mol lit^{-1} sec^{-1}.

Time for completion = $\frac{C_0}{k}$

$t_{1/2}$ (half life period) at $t_{1/2}$, $C_t = \frac{C_0}{2}$,

so $kt_{1/2} = \frac{C_0}{2} \Rightarrow$ **$t_{1/2} = \frac{C_0}{2k}$**

$\boxed{\therefore t_{1/2} \propto C_0}$

Examples of zero order reactions:

Generally decomposition of gases on metal surfaces at high concentrations follow zero order kinetics.

$2PH_3(g) \xrightarrow{Ni} 2P + 3H_2$ Rate = $K[PH_3]^o$ $2HI(g) \xrightarrow{Au} H_2 + I_2$

$2NH_3(g) \xrightarrow{N} N_2 + 3H_2$

$H_2 + Cl_2 \xrightarrow{hv} 2HCl$ Rate = $R[H_2]^o[Cl_2]^o$

(ii) First order reaction

(i) Let a 1st order reaction is

	A	⟶	Products	
conc.	'a'		0	t = 0
	'a–x'		.....	t = 't'

Let $\frac{dx}{dt}$ be the rate of reaction at time 't'

$\therefore \quad \frac{dx}{dt} = k(a-x)_1$ or $\frac{dx}{a-x} = kdt.$

On solving $t = \frac{2.303}{k} \log \frac{a}{a-x}$

$$k = \frac{2.303}{t} \log \frac{C_0}{C_t}$$

$$\boxed{k = \frac{2.303}{t} \log \frac{C_0}{C_t}}$$

Wilhemy formula:
$$C_t = C_0 e^{-kt}$$

Interval formula:
$$k = \frac{2.3032}{(t_2 - t_1)} \log \frac{C_1}{C_2}$$

If any substance is growing/increasing following first order kinetics then:

$$k = \frac{2.303}{t} \log\left(\frac{a+x}{a}\right)$$

where a is initial concentration of the substance and x is the increment in its concentration after time t.

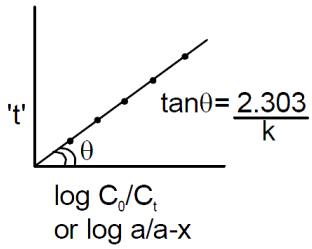
Half life time ($t_{1/2}$) $k = \frac{2.303}{t} \log \frac{2C_0}{C_0}$

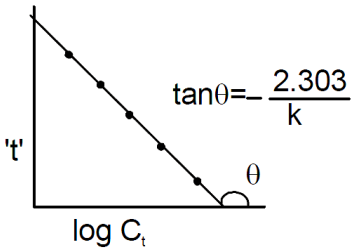
$$\Rightarrow \quad t_{1/2} = \frac{2.303 \log 2}{k} = \frac{0.693}{k}$$

$\therefore$ Half life period for a 1st order reaction is a constant quantity.

Graphical Representation:

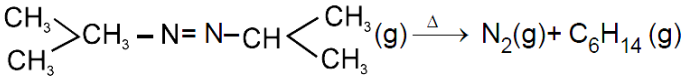
$$t = -\frac{2.303}{k} \log C_t + \frac{2.303}{R} \log C_0$$



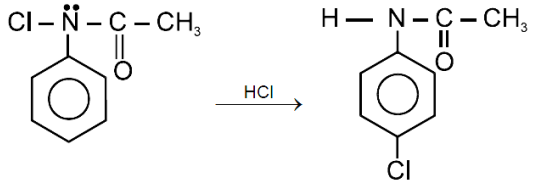


Examples of 1st order reactions:

1. Decomposition of azoisopropane



2. Conversion of N–chloro acetanilide into p–chloroacetanilide



3. $H_2O_2 \longrightarrow H_2O + \frac{1}{2}O_2$

4. $NH_4NO_2 \longrightarrow 2H_2O + N_2$

5. Radioactive decay

- All radioactive decays are always first order kinetics.

$^{226}_{88}Ra \longrightarrow {}^{222}_{86}Rn + {}_2He^4$

Solved Examples

Example: Calculate $\frac{t_{0.75}}{t_{0.50}}$ for a 1st order reaction

Solution: $k = \frac{2.303}{t_{1/4}}\log\frac{C_0}{\frac{1}{4}C_0} = \frac{2.303}{t_{1/2}}\log\frac{C_0}{\frac{V_0}{2}} \Rightarrow \frac{t_{1/4}}{t_{1/2}} = \frac{\log 4}{\log 2} = \frac{2\log 2}{\log 2} = 2$

Example: At least how many half-lives should elapse for a 1st order reaction A $\longrightarrow$ products so that the reaction is at least 95% completed? (log 2 = 0.3)

(A) 4 (B) 5 (C) 6 (D) 7

Solution: (B) $100 \xrightarrow{t_{1/2}} 50 \xrightarrow{t_{1/2}} 25 \xrightarrow{t_{1/2}} 12.5 \xrightarrow{t_{1/2}} 6.25 \xrightarrow{t_{1/2}} 3.125$

0% 50% 75% 87.5% 93.75% 96.875%

Pseudo first order reaction:

The order of a reaction is sometimes altered by conditions. Consider a chemical reaction between two substances when one reactant is present in large excess. During the hydrolysis of 0.01 mol of ethyl acetate with 10 mol of water, amounts of the various constituents at the beginning (t = 0) and completion (t) of the reaction are given as under.

$CH_3COOC_2H_5 + H_2O \xrightarrow{H^+} CH_3COOH + C_2H_5OH$

t = 0	0.01 mol	10 mol	0 mol	0 mol
t	0 mol	9.9 mol	0.01 mol	0.01 mol

The concentration of water does not get altered much during the course of the reaction. So, in the rate equation

Rate = k' $[CH_3COOC_2H_5]$ $[H_2O]$

the term $[H_2O]$ can be taken as constant. The equation, thus, becomes

Rate = k $[CH_3COOC_2H_5]$

where k = k' $[H_2O]$ and the reaction behaves as first order reaction. Such reactions are called pseudo first order reactions.

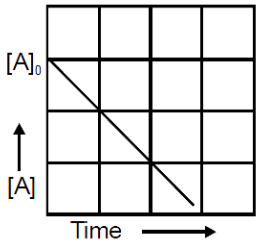
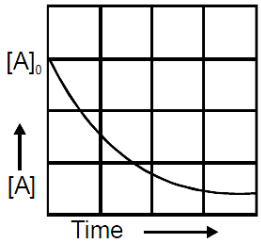
Inversion of cane sugar is another pseudo first order reaction.

$C_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+} C_6H_{12}O_6 + C_6H_{12}O_6$

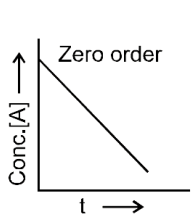
Cane sugar Glucose Fructose

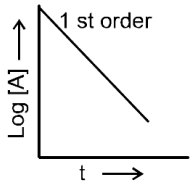
Rate = k $[C_{12}H_{22}O_{11}]$

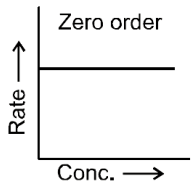
Table: Characteristics of First-and Second-Order Reactions of the Type A ⟶ Products

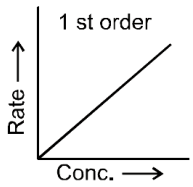
	Zero Order	First-Order
Differential Rate law	$\frac{-\Delta A}{\Delta t} = k[A]^{\circ}$	$-\frac{\Delta[A]}{\Delta t} = k[A]$
(Integrated Rate law)	$[A]_t = [A]_0 - kt$	$\ln [A]_t = -kt + \ln [A]_0$
Linear graph	$[A]_t$ versus t	ln [A] versus t
Half-life	$t_{1/2} = \frac{[A]_0}{2k}$	$t_{1/2} = \frac{0.693}{k}$
	(depends on $[A]_0$)	(independent of $[A]_0$)
		

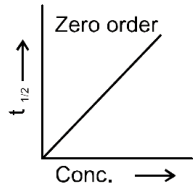
Graphical comparison of different orders

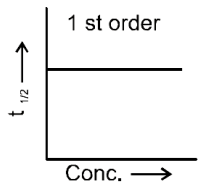












(b) Methods to determine order of reaction

(i) Initial rate method:

- By comparison of different initial rates of a reaction by varying the concentration of one of the reactants while others are kept constant

  $r = k [A]^a [B]^b [C]^c$ if [B] = constant
  [C] = constant

  then for two different initial concentrations of A we have

$$r_{0_1} = k[A_0]_1^a \; r_{0_2} = k[A_0]_2^a \Rightarrow \frac{r_{0_1}}{r_{0_2}} = \left(\frac{[A_0]_1}{[A_0]_2}\right)^x$$

or in log form we have $a = \dfrac{\log\left(r_{0_1} / r_{0_2}\right)}{\log\left([A_0]_1 / [A_0]_2\right)}$

Solved Examples

Example: The pressure of a gas decomposing at the surface of a solid catalyst has been measured at different times and the results are given below:

t/s	0	100	200	300
p/Pa	4.00×10^3	3.50×10^3	3.00×10^3	2.5×10^3

Determine the order of reaction, its rate constant and half-life period.

Solution: It Can be seen that rate of reaction between different time intervals is:

$$0\text{–}100 \text{ s, rate} = \frac{[3.50 - 4.00] \times 10^3 \text{ Pa}}{100} = 5 \text{ Pa/s}$$

$$100\text{–}200\text{s, rate} = \frac{[3.00 - 3.50] \times 10^3 \text{ Pa}}{100\text{s}} = 5 \text{ Pa/s}$$

$$200\text{–}300\text{s, rate} = -\frac{[2.50 - 3.00] \times 10^3 \text{ Pa}}{100} = 5 \text{ Pa/s}$$

We notice that the rate remains constant and therefore, reaction is of zero order. Alternatively, if we plot a p against t, it is a straight line again indicating it is a zero order reaction.

k = rate = 5 Pa/s

$$t_{1/2} = \frac{\text{initial concentration or pressure}}{2k} = \frac{4.00 \times 10^3 \text{ Pa}}{2 \times 5 \text{ Pa s}^{-1}} = 400\text{s}$$

(ii) Method of half lives:

- The half lives of each order is unique so by comparing half lives we can determine order

for n^{th} order reaction $\quad t_{1/2} \propto \dfrac{1}{[R_0]^{n-1}}$

$$\frac{t_{1/2}}{t'_{1/2}} = \frac{(R'_0)^{n-1}}{(R_0)^{n-1}}$$

Solved Examples

Example: In the reduction of nitric gas with hydrogen, the reaction was found to be 50% complete in 210 seconds when the initial pressure of the mixture was 200 mm. In a second experiment the time of half reaction was 140 seconds when the initial pressure was 300 mm. Calculate the total order of the reaction.

Solution: For a nth order reaction ($n \neq 1$), $t_{1/2} \propto \frac{1}{c_0^{n-1}}$

$$\frac{210}{140} = \left(\frac{300}{200}\right)^{n-1} \quad n = 2$$

ARHENIUS THEORY OF REACTION RATE

Most of the chemical reactions are accelerated by increase in temperature. For example, in decomposition of N_2O_5, the time taken for half of the original amount of material to decompose is 12 min at 50ºC, 5 h at 25ºC and 10 days at 0ºC. You also know that in a mixture of potassium permanganate ($KMnO_4$) and oxalic acid ($H_2C_2O_4$), potassium permanganate gets decolourised faster at a higher temperature than that at a lower temperature.

It has been found that for a chemical reaction with rise in temperature by 10°, the rate constant is nearly doubled.

The temperature dependence of the rate of a chemical reaction can be accurately explained by Arrhenius equation (4.18). It was first proposed by Dutch chemist, J.H. van't Hoff but Swedish chemist, Arrhenius provided its physical justification and interpretation. k = A e -Ea /RT (4.18) where A is the Arrhenius factor or the frequency factor. It is also called pre-exponential factor. It is a constant specific to a particular reaction. R is gas constant and Ea is activation energy measured in joules/mole (J mol –1).

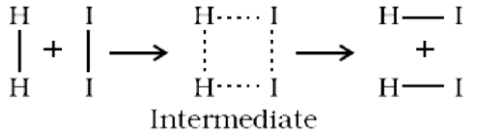


Fig. Formation of HI through the intermediate Intermediate

It can be understood clearly using the following simple reaction

$$H_2(g) + I_2(g) \rightarrow 2HI(g)$$

According to Arrhenius, this reaction can take place only when a molecule of hydrogen and a molecule of iodine collide to form an unstable intermediate (Fig.). It exists for a very short time and then breaks up to form two molecules of hydrogen iodide.

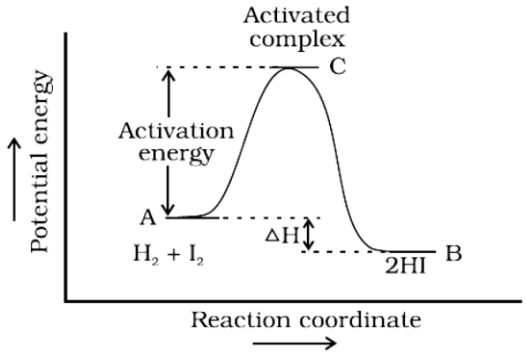
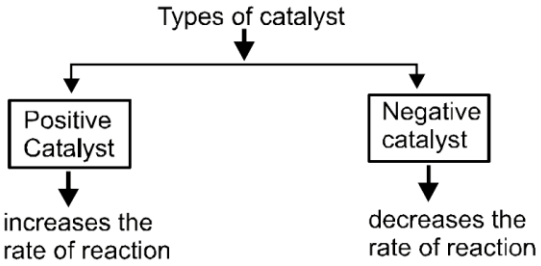


Diagram showing plot of potential energy vs reaction coordinate

The energy required to form this intermediate, called activated complex (C), is known as activation energy (Ea). This diagram is obtained by plotting potential energy vs reaction coordinate. Reaction coordinate represents the profile of energy change when reactants change into products. Some energy is released when the complex decomposes to form products. So, the final heat of the reaction depends upon the nature of reactants and products.

CATALYST AND CATALYSIS

A **catalyst** is a substance, which increases the rate of a reaction without itself being consumed at the end of the reaction, and the phenomenon is called **catalysis**.



Catalyst are generally foreign substances but sometimes one of the product may act as a catalyst and such catalyst is called **"auto catalyst"** and the phenomena is called **auto catalysis**.

Examples of catalysis

(a) Thermal decomposition of $KClO_3$ is found to be accelerated by the presence of MnO_2. Here MnO_2 (foreign substance) acts as a catalyst.

$$2KClO_3 + [MnO_2] \rightarrow 2KCl + 3O_2\uparrow + [MnO_2]$$

- MnO_2 can be received in the same composition and mass at the end of the reaction.

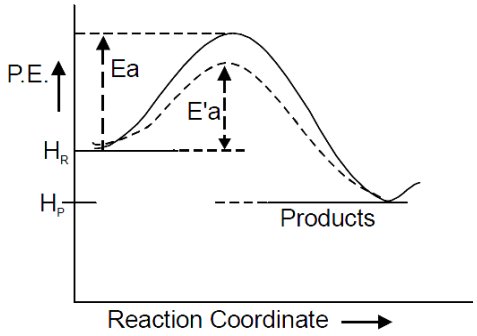
(b) In the permanganate titration of oxalic acid initially there is slow discharge of the colour of permanganate solution but afterwards the discharge of the colour become faster. This is due to the formation of $MnSO_4$ during the reaction which acts as a catalyst for the same reaction. Thus, $MnSO_4$ is an **"auto catalyst"** for this reaction.

This is an example of auto catalyst.

$$2KMnO_4 + 3H_2SO_4 + 5H_2C_2O_2 \rightarrow K_2SO_4 + 8H_2O + 10CO_2$$

(a) General characteristics of catalyst:

- A catalyst does not initiate the reaction. It simply fastens it.
- Only a small amount of catalyst can catalyse the reaction.
- A catalyst does not alter the position of equilibrium i.e. magnitude of equilibrium constant and hence ΔG^o. It simply lowers the time needed to attain equilibrium. This means if a reversible reaction in absence of catalyst completes to go to the extent of 75% till attainment of equilibrium, and this state of equilibrium is attained in 20 minutes then in presence of a catalyst also the reaction will go to 75% of completion before the attainment of equilibrium but the time needed for this will be less than 20 minutes.
- A catalyst drives the reaction through a low energy path and hence E_a is less. That is, the function of the catalyst is to lower down the activation energy.



E_a = Energy of activation in absence of catalyst.

E'_a = Energy of activation in presence of catalyst.

$E_a - E'_a$ = lowering of activation energy by catalyst.

(b) Molecularity and Order:

The number of molecules that react in an elementary step is the molecularity of the elementary reaction. Molecularity is defined only for the elementary reactions and not for complex reactions. No elementary reactions involving more than three molecules are known, because of very low probability of near-simultaneous collision of more than three molecules.

The rate law for the elementary reaction

$aA + bB \longrightarrow$ products rate = $k[A]^a[B]^b$,

where a + b = 1, 2 or 3.

For an elementary reaction, the orders in the rate law equal the coefficients of the reactants.

While, the order is defined for complex as well as elementary reactions and is always experimentally calculated by the mechanism of the reaction, usually by the slowest step of the mechanism known as **rate determining step of the reaction.**

Comparison between Molecularity and order of reaction

S.No.	Molecularity of Reaction	Order of Reaction
1.	It is defined as the no. of molecules of reactant taking part in a chemical reaction eq $NH_4NO_2 \rightarrow N_2 + 2\ H_2O$ m = 1	It is defined as the sum of the power of concentration terms that appear in rate law. $NH_4NO_2 \rightarrow N_2 + 2\ H_2O$. Rate= K $[NH_4NO_2]$
2.	It is always a whole number It can neither be zero nor fractional.	It may be zero, fractional or integer.
3.	It is derived from RDS in the mechanism of reaction.	It is derived from rate expression.
4.	It is theoretical value.	It is experimental value.
5.	Reactions with molecularity 4 are rare.	Reactions with O.R are also rare.
6.	Molecularity is in independent of Pressure and temperature.	OR. depends upon pressure and temperature.

EXERCISE-1

1. $xA + yB \rightarrow zC$.

If $-\frac{d[A]}{dt} = -\frac{d[B]}{dt} = 1.5\frac{d[C]}{dt}$ = then x, y and z can be:

(A) 1,1,1 (B) 3,2,3

(C) 3,3,2 (D) 2,2,3

2. The rate of a reaction is expressed in different ways as follows ;

$+ 1/2(d[C]/dt) = - 1/3 (d[D]/dt) = + 1/4 (d[A]/dt) = - (d[B]/dt)$ The reaction can be:

(A) $4\ A + B \longrightarrow 2C + 3D$

(B) $B + 3D \longrightarrow 4A + 2C$

(C) $4A + 2B \longrightarrow 2C + 3D$

(D) $B + (1/2)\ D \longrightarrow 4A + 3C$

3. Consider the chemical reaction:

$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$

The rate of this reaction can be expressed ; in terms of time and of concentration of $N_2(g)$, $H_2(g)$ or $NH_3(g)$. Identify the correct relationship amongst the rate expressions.

(A) $\text{Rate} = -\frac{d[N_2]}{dt} = -\frac{1}{3}\frac{d[H_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$

(B) $\text{Rate} = -\frac{d[N_2]}{dt} = -\frac{3d[H_2]}{dt} = \frac{2d[NH_3]}{dt}$

(C) $\text{Rate} = -\frac{d[N_2]}{dt} = -\frac{1}{3}\frac{d[H_2]}{dt} = \frac{d[NH_3]}{dt}$

(D) $\text{Rate} = -\frac{d[N_2]}{dt} = \frac{d[H_2]}{dt} = \frac{d[NH_3]}{dt}$

4. In a reaction $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$ the rate of appearance of NH_3 is 2.5×10^{-4} mol L^{-1} sec^{-1}. The rate of reaction & rate of disappearance of H_2 will be (in mol L^{-1} sec^{-1})

(A) 3.75×10^{-4}, 1.25×10^{-4}

(B) 1.25×10^{-4}, 2.5×10^{-4}

(C) 1.25×10^{-4}, 3.75×10^{-4}

(D) 5.0×10^{-4}, 3.75×10^{-4}

5. $3A \rightarrow 2B$, rate of reaction $\frac{+d[B]}{dt}$ is equal to:

(A) $-\frac{3}{2}\frac{d[A]}{dt}$ (B) $-\frac{2}{3}\frac{d[A]}{dt}$

(C) $-\frac{1}{3}\frac{d[A]}{dt}$ (D) $+2\frac{d[A]}{dt}$

6. In the reaction; $A + 2B \longrightarrow 3C + D$, which of the following expression does not describe changes in the concentration of various species as a function of time:

(A) $\{d[C]/dt\} = -\{3d[A]/dt\}$

(B) $\{3d[D]/dt\} = \{d[C]/dt\}$

(C) $\{3d[B]/dt\} = -\{2d[C]/dt\}$

(D) $\{2d[B]/dt\} = -\{d[A]/dt\}$

7. The rate constant of reaction $2\ A + B \rightarrow C$ is 2.57×10^{-5}. It $mole^{-1}$ sec^{-1} after 10 sec. 2.65×10^{-5}. It $mole^{-1}$ sec^{-1} after 20 sec. and 2.55×10^{-5}. It $mole^{-1}$ sec^{-1} after 30 sec. The order of the reaction is:

(A) 0 (B) 1

(C) 2 (D) 3

8. For a first order reaction, the plot of 't' against log C gives a straight line with slope equal to:

(A) (k / 2.303) (B) (– k / 2.303)

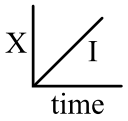
(C) (ln k / 2.303) (D) – k.

9. In a first order reaction the amount of reactant decayed in three half lives (let a be is initial amount) would be:

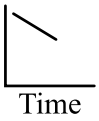
(A) 7a / 8 (B) a / 8

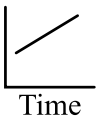
(C) a / 6 (D) 5a / 6

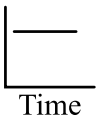
10. Graph between concentration of the product and time of the reaction A → B is of the type

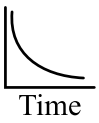


Hence graph between – d[A]/dt and time will be of the type:

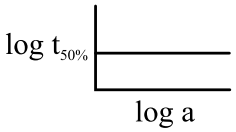
(A) 

(B) 

(C) 

(D) 

11. A graph plotted between log $t_{50\%}$ vs. log concentration is a straight line. What conclusion can you draw from this graph.



(A) $n = 1; t_{1/2} \propto a$

(B) $n = 2, t_{1/2} \propto 1/a$

(C) $n = 1; t_{1/2} = (0.693 / k)$

(D) None of these

12. K for a zero order reaction is 2×10^{-2} mol L^{-1} sec^{-1}. If the concentration of the reactant after 25 sec is 0.5 M, the initial concentration must have been.

(A) 0.5 M (B) 1.25 M

(C) 12.5 M (D) 1.0 M

13. Plot of log(a–x) vs time t is straight line. This indicates that the reaction is of

(A) Second order (B) First order

(C) Zero order (D) Third order

14. In a first order reaction the concentration of the reactant is decreased from 1.0 M to 0.25 M in 20 min. The rate constant of the reaction would be –

(A) 10 min^{-1} (B) 6.931 min^{-1}

(C) 0.6931 min^{-1} (D) 0.06931 min^{-1}

15. In a certain reaction, 10% of the reactant decomposes in one hour, 20 % in two hours, 30% in three hours and so on the dimensions of the rate constant is:

(A) $hour^{-1}$ (B) mole $litre^{-1}$ sec^{-1}

(C) litre $mole^{-1}$ sec^{-1} (D) mole sec^{-1}

16. If a first order reaction is completed to the extent of 75% and 50% in time intervals, t_1 and t_2, what is the ratio, $t_{1:}$ $t_{2?}$

(A) ln 2 (B) $\frac{\ln(3/4)}{\ln 2}$

(C) 2 (D) 1/2

17. Rate of which reactions increases with temperature:

(A) of any

(B) of exothermic reactions

(C) of endothermic

(D) of none.

18. For a zero order reaction. Which of the following statement is false:

(A) the rate is independent of the temperature of the reaction.

(B) the rate is independent of the concentration of the reactants.

(C) the half life depends as the concentration of the reactants.

(D) the rate constant has the unit mole It^{-1} sec^{-1}.

19. A large increase in the rate of a reaction for a rise in temperature is due to

(A) increase in the number of collisions

(B) the increase in the number of activated molecules

(C) The shortening of mean free path

(D) the lowering of activation energy

20. The rate constant, the activation energy and the frequency factor of a chemical reaction at 25°C are $3.0 \times 10^{-4}\ s^{-1}$, 104.4 KJ mol^{-1} and $6.0 \times 10^{14}\ s^{-1}$ respectively. The value of the rate constant as $T \to \infty$ is:

(A) $2.0 \times 10^{18}\ s^{-1}$ (B) $6.0 \times 10^{14}\ s^{-1}$

(C) ∞ (D) $3.6 \times 10^{30}\ s^{-1}$

21. According to collision theory of reaction rates –

(A) Every collision between reactant leads to chemical reaction

(B) Rate of reaction is proportional to velocity of molecules

(C) All reactions which occur in gaseous phase are zero order reaction

(D) Rate of reaction is directly proportional to collision frequency.

22. In the following reaction:

$xA \longrightarrow yB$

$$\log\left[-\frac{d[A]}{dt}\right] = \log\left[\frac{d[B]}{dt}\right] + \log 2$$

where –ve sign indicates rate of disappearance of the reactant. Thus, x: y is:

(A) 1: 2 (B) 2: 1

(C) 3: 1 (D) 3: 10

23. $aA + bB \longrightarrow$ Product, $dx/dt = k\,[A]^a\,[B]^b$. If conc of A is doubled, rate is four times. If conc. of B is made four times, rate is doubled. What is relation between rate of disappearance of A and that of B?

(A) $\{-d[A]/dt\} = \{-d[B]/dtK$

(B) $-\{d[A]/dt\} = -4\{d[B]/dt\}$

(C) $-4\{d[A]/dt) = \{-d[B]/dt\}$

(D) None of these

24. For a reaction $pA + qB \to$ products, the rate law expression is $r = k[A]^1\,[B]^m$, then:

(A) $(p+1) < (1+m)$

(B) $(p+q) > (1+m)$

(C) $(p+q)$ may or may not be equal to $(1+m)$

(D) $(p+q) = (1+m)$

25. For the reaction $H_2 + Br_2 \to 2$ HBr overall order is found to be 3/2. The rate of reaction can be expressed as:

(A) $[H_2][Br_2]^{1/2}$ (B) $[H_2]^{1/2}\,[Br_2]$

(C) $[H_2]^{3/2}\,[Br_2]^0$ (D) All of these

26. If concentration of reactants is increased by a factor x then the rate constant k becomes:

(A) $\ln \frac{k}{x}$ (B) $\frac{k}{x}$

(C) $k + x$ (D) k

27. The rate of certain hypothetical reaction $A + B + C \to$ products is given by $r = \frac{-d[A]}{dt} = K\,[A]^{1/2}[B]^{1/3}\,[C]^{1/4}$. The order of the reaction:

(A) 1 (B) $\frac{1}{2}$

(C) 2 (D) $\frac{13}{12}$

28. $2A \to B + C$ it would be a zero order reaction when:

(A) The rate of reaction is proportional to square of conc. of A

(B) The rate of reaction remains same at any conc. of A

(C) The rate remains unchanged at any conc. of B and C

(D) The rate of reaction doubles if conc. of B is increased to double

29. The rate equation for the reaction $2A + B \to C$ is found to be: rate k[A][B]. The correct statement in relation to this reaction is that the

(A) Rate of formation of C is twice the rate of disappearance of A.

(B) Half life is a constant

(C) Unit of k must be s^{-1}

(D) Value of k is independent of the initial concentration of A and B

30. The rate constant for a reaction is 10.8×10^{-5} mol L^{-1} S^{-1}. The reaction obeys:

(A) First order (B) Zero order

(C) Second order (D) All are wrong

31. For the reaction, $2NO(g) + 2H_2(g) \longrightarrow N_2(g) + 2H_2O(g)$ the rate expression can be written in the following ways:

$\{dt\ [N_2] / dt\} = k_1\ [NO][H_2]$;

$\{d[H_2O] / dt) = k[NO][H_2]$;

$\{- d[NO] / dt\} = k'_1\ [NO]\ [H_2]$;

$\{-d[H_2] / dt\} = k''_1\ [NO][H_2]$

The relationship between k, k_1, k'_1 and k''_1. is:

(A) $k = k_1 = k'_1 = k''_1$

(B) $k = 2k_1 = k'_1 = k''_1$

(C) $k = 2k'_1 = k_1 = k''_1$

(D) $k = k_1 = k'_1 = 2\ k''_1$

32. If a reaction gets completed in finite time then its order can be:

(A) 3 (B) 2

(C) 1 (D) Zero

33. For the irreversible process, A + B $\longrightarrow$ products, the rate is first–order w.r.t. A and second–order w.r.t. B. If 1.0 mol each of A and B introduced into a 1.0 L vessel, and the initial rate was 1.0×10^{-2} mol L^{-1} s^{-1}, rate when half reactants have been turned into products is:

(A) 1.25×10^{-3} mol L^{-1} s^{-1}

(B) 1.0×10^{-2} mol L^{-1} s^{-1}

(C) 2.50×10^{-3} mol L^{-1} s^{-1}

(D) 2.0×10^{-2} mol L^{-1} s^{-1}

34. The rate law for the single step reaction $2A + B \rightarrow 2C$, is given by

(A) Rate = K [A][B]

(B) Rate = K $[A]^2[B]$

(C) Rate = K [2A][B]

(D) Rate = K$[A]^2[B]^0$

35. For the reaction A + B $\rightarrow$ Products, it is found that the order of A is 2 and of B is 3 in the rate expression. When concentration of both is doubled the rate will increase by a factor of:

(A) 10 (B) 6

(C) 32 (D) 16

36. For a reaction A + B $\rightarrow$ products, the rate of the reaction was doubled when the concentration of. A was doubled, the rate was again doubled when the conc. of A & B were doubled the order of the reaction with respect to A & B are:

(A) 1,1 (B) 2,0

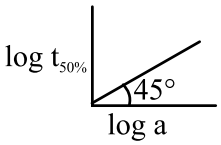
(C) 1,0 (D) 0,1

37. The rate constant for the reaction A $\rightarrow$ B is 2×10^{-4} It. mol^{-1} min^{-1}. The concentration of A at which rate of the reaction is $(1/12) \times 10^{-5}$ M sec^{-1} is:

(A) 0.25 M (B) $(1/20)\ \sqrt{5/3}$ M

(C) 0.5 M (D) None of these

38. What will be the order of reaction and rate constant for a chemical change having log $t_{50\%}$ vs log concentration of (A) curves as:



(A) 0, 1/2 (B) 1, 1

(C) 2, 2 (D) 3, 1

39. The rate constant for a reaction is $\frac{\ln 2}{10}$ min^{-1}. What will be the order of reaction and time taken to change concentration from 1 M to 0.25 M.

(A) one, 10 min (B) zero, 10 min

(C) one, 20 min (D) two, 20 min

40. In the first order reaction 75% of the reactant disappeared in 1.388 hrs. Calculate the rate constant of the reaction:

(A) 1 s^{-1} (B) 2.8×10^{-4} s^{-1}

(C) 17.2×10^{-3} s^{-1} (D) 1.8×10^{-3} s^{-1}

41. In the case of zero order reaction, the ratio of time required for 75% completion to 50% completion is:

(A) ln 2 (B) 2

(C) 1.5 (D) None

42. In a I order reaction A $\longrightarrow$ products, the concentration of the reactant decreases to 6.25% of its initial value in 80 minutes. What is (i) the rate constant and (ii) the rate of the reaction, 100 minutes after the start, if the initial concentration is 0.2 mole/litre?

(A) 2.17×10^{-2} min^{-1}, 3.47×10^{-4} $mol.litre^{-1}$ min^{-1}

(B) 3.465×10^{-2} min^{-1}, 2.166×10^{-4} $mol.litre^{-1}$ min^{-1}

(C) 3.465×10^{-3} min^{-1}, 2.17×10^{-3} $mol.litre^{-1}$ min^{-1}

(D) 2.166×10^{-3} min^{-1}, 2.667×10^{-4} $mol.litre^{-1}$ min^{-1}

43. The data for the reaction A + B $\rightarrow$ C is

Exp.	$[A]_0$	$[B]_0$	initial Rate
1	0.012	0.035	0.10
2	0.024	0.035	0.80
3	0.012	0.070	0.10
4	0.024	0.070	0.80

(A) $r = k[B]^3$ (B) $r = k[A]^3$

(C) $r = k[A][B]^4$ (D) $r = k[A]^2[B]^2$

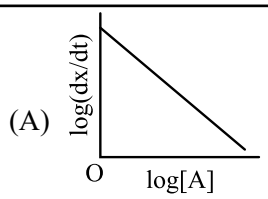
44. A + B $\longrightarrow$ Product, $\frac{dx}{dt} = k[A]^a[B]^b$

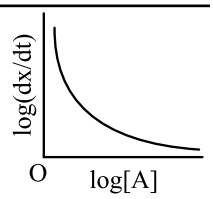
If $\left(\frac{dx}{dt}\right) = k$, then order is:

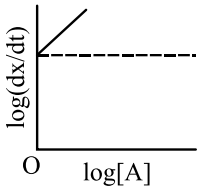
(A) 4 (B) 2

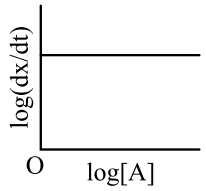
(C) 1 (D) 0

45. A $\rightarrow$ Product and $\left(\frac{dx}{dt}\right) = k[A]^2$. If $\log\left(\frac{dx}{dt}\right)$ is plotted against log [A], then graph is of the type:

(A) 

(B) 

(C) 

(D) 

46. For the reaction A $\longrightarrow$ Products, $-\frac{d[A]}{dt} = k$ and at different time interval, [A] values are:

Time	0	5 min	10 min	15 min
[A]	20 mol	18 mol	16 mol	14 mol

At 20 minute, rate will be:

(A) 12 mol /min (B) 10 mol/min

(C) 8 mol/min (D) 0.4 mol/min

47. The rate law for a reaction A + B $\rightarrow$ product is rate = $K[A]^1[B]^2$. Then which one of the following statement is false:

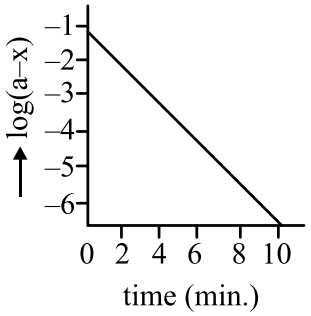
(A) If [B] is held constant while [A] is doubled, the reaction will proceed twice as fast

(B) If [A] is held constant while [B] is reduced to one quarter, the rate will be halved

(C) If [A] and [B] are both doubled, the reaction will proceed 8 times as fast

(D) This is a third order reaction

48. The conversion of vinyl allyl ether to pent-4-enol follows a certain kinetics. The following plot is obtained for such a reaction.



The order for the reaction is

(A) zero (B) –1

(C) 1 (D) 2

49. The reaction A(s) $\rightarrow$ 2 B(g) + C(g) is first order. The pressure after 20 min. and after very long time are 150 mm Hg and 225 mm Hg. The value of rate constant and pressure after 40 min. are:
(A) 0.05 in 1.5 min^{-1}, 200 mm
(B) 0.5 in 2 min^{-1}, 300 mm
(C) 0.05 in 3 min^{-1}, 300 mm
(D) 0.05 in 3 min^{-1}, 200 mm

50. Activation energy of a reaction is –
(A) The energy released during the reaction
(B) The energy evolved when activated complex is formed
(C) Minimum amount of energy needed to overcome the potential barrier of reaction
(D) The energy needed to form one mole of the product.

51. The minimum energy for molecules to enter into chemical reaction is called.
(A) Kinetic energy
(B) Potential energy
(C) Threshold energy
(D) Activation energy

52. The activation energy for the forward reaction X $\rightarrow$ Y is 60 KJ mol^{-1} and ΔH is –20 KJ mol^{-1}. The activation energy for the backward reaction Y $\rightarrow$ X is:
(A) 80 KJ mol^{-1} (B) 40 KJ mol^{-1}
(C) 60 KJ mol^{-1} (D) 20 KJ mol^{-1}

53. For producing the effective collisions, the colliding molecules must posses:
(A) A certain minimum amount of energy
(B) Energy equal to or greater than threshold energy
(C) Proper orientation
(D) Threshold energy as well as proper orientation of collision.

54. In a gaseous state reaction, A_2 (g) $\longrightarrow$ B(g) + (1/2)C (g), The increase in pressure from 100 mm to 120 mm is notices in 5 minutes. The average rate of disappearance of A_2 in mm min^{-1} is:
(A) 4 (B) 8
(C) 16 (D 2

EXERCISE-2

1. The decomposition of NH_3 on platinum surface, $2\,NH_3\,(g) \xrightarrow{Pt} N_2\,(g) + 3H_2(g)$, is zero order with $k = 2.5 \times 10^{-4}$ mol L^{-1} s^{-1}. What are the rates of production of N_2 and H_2?

2. The rate of formation of a dimer in a second order dimerisation reaction is 9.5×10^{-5} mol L^{-1} s^{-1} at 0.01 mol L^{-1} monomer concentration. Calculate the rate constant.

3. The half life period of a substance is 50 minutes at a certain initial concentration. When the concentration is reduce to one half of its initial concentration, the half life period is found to be 25 minutes. Calculate the order of reaction.

4. The decomposition of N_2O_5 in CCl_4 solution follows the first order rate law. The concentrations of N_2O_5 measured at different intervals are given below:

Time in seconds(t)	0	80	160	410	600	1130	1720
$[N_2O_5]$mol /L	5.5	5.0	4.8	4.0	3.4	2.4	1.6

Calculate the rate constant at t = 410 s and t = 1130 s. What do these results show?

5. Show that in case of a first order reaction, the time required for 99.9% of the reaction to take place is about ten times than that required for half the reaction.

NSEJS PYQ

1. For the reaction, $2A + B \rightarrow C$ which relationship is correct? **[NSEJS-2014]**

(A) $\Delta[A] = \Delta[C]$ (B) $-\Delta[A] = \Delta[C]$

(C) $-2\Delta[A] = \Delta[C]$ (D) $-\Delta[A] = 2\Delta[C]$

2. For a first order reaction, the ratio of the times taken for completion of 99.9% and 50% of the reaction is **[NSEJS-2015]**

(A) 8 (B) 9

(C) 12 (D) 10

3. The following data was recorded for the reaction $A + B \rightarrow$ Product at 298 K.

[NSEJS-2015]

Experiment No.	[A]	[B]	Rate of reaction
1	1.00 M	0.15 M	4.20×10^{-3}
2	2.00 M	0.15 M	8.40×10^{-3}
3	1.00 M	0.30 M	8.40×10^{-3}

From the above data one can conclude that

(A) Rate $\propto [A]^2[B]$ (B) Rate $\propto [A][B]^2$

(C) Rate $\propto [A]^2[B]^2$ (D) Rate $\propto [A][B]$

4. Decay of organic matter such as vegetation of chemical reaction with sulphur containing mineral in rock and soil produces gas which has rotten egg smell. Calculate the percentage dissociation of the gas produced by the sulphur reducing bacteria if dissociation constant (Kc) of the gas is 1.0×10^{-6}. There are 0.1 moles of the gas in 1.6 dm_3 vessel **[NSEJS-2015]**

(i) Write the equation for dissociation of the gas produced in the reaction.

(ii) If x moles of the gas are dissociated, write the expression for equilibrium constant in terms of x. (Detail working expected)

(iii) Calculate the value of x and hence percentage dissociation.

5. In the conversion of compound X into compound Z, it was found that the reaction proceeded by way of Y, which could be isolated. The following steps were involved.

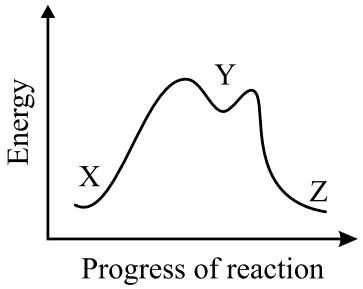
[NSEJS-2016]

$X \rightarrow Y$; endothermic process

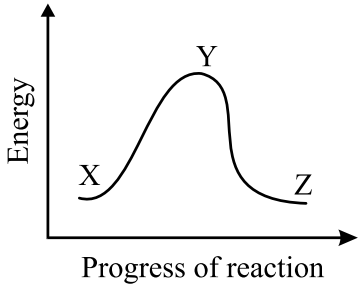
$Y \rightarrow Z$; exothermic process

Which is the reaction profile of this conversion from X to Z?

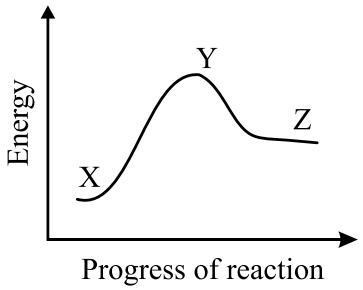
(A)



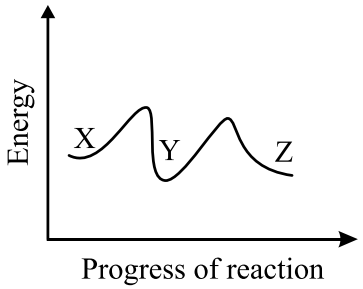
(B)



(C)



(D)



Answer Key

EXERCISE-1

1. (C)	**2.** (B)	**3.** (A)	**4.** (C)
5. (B)	**6.** (D)	**7.** (C)	**8.** (B)
9. (A)	**10.** (C)	**11.** (C)	**12.** (D)
13. (B)	**14.** (D)	**15.** (B)	**16.** (C)
17. (A)	**18.** (A)	**19.** (B)	**20.** (B)
21. (D)	**22.** (B)	**23.** (B)	**24.** (C)
25. (D)	**26.** (D)	**27.** (D)	**28.** (B)
29. (D)	**30.** (B)	**31.** (B)	**32.** (D)
33. (A)	**34.** (B)	**35.** (C)	**36.** (C)
37. (C)	**38.** (A)	**39.** (C)	**40.** (B)
41. (C)	**42.** (B)	**43.** (B)	**44.** (D)
45. (C)	**46.** (D)	**47.** (B)	**48.** (C)
49. (D)	**50.** (C)	**51.** (C)	**52.** (A)
53. (D)	**54.** (B)		

EXERCISE-2

1. $\therefore$ Rate of production of $N_2 = \frac{d[N_2]}{dt} = 2.5 \times 10^{-4}\ mol\ L^{-1}\ s^{-1}$

Rate of production of $H_2 = \frac{d[H_2]}{dt} = 3 \times (2.5 \times 10^{-4}\ mol\ L^{-1}\ s^{-1})\ 7.5 \times 10^{-4}\ mol\ L^{-1}\ s^{-1}$

2. $k = 0.95\ L\ mol\ L^{-1}\ s^{-1}$

3. $1 - n = 1$ or $n = 0$. Hence, the reaction is of zero-order.

4. These result show that the rate constant is nearly constant at different times.

5. $\frac{t_{99.9\%}}{t_{1/2}} = \frac{3}{0.3010} = 10$

NSEJS PYQ

1. (D) **2.** (D) **3.** (D)

4. % dissociation $= \frac{x}{0.1} \times 100 = 0.4\%$ **5.** (A)